



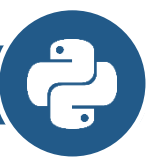
Python

Conditional Statements

#PythonNotes

Indentation in Python

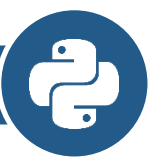
- In Python, indentation is used to declare a block. If two statements are at the same indentation level, then they are the part of the same block.
- For the ease of programming and to achieve simplicity, python doesn't allow the use of curly braces or parentheses for the block level code.
- Indentation is the most used part of the python programming language.
- Generally, a tab space or four spaces are given to indent the statements in python



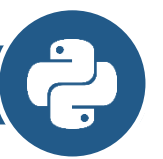
Conditional Statements



- Conditional Statements performs different computations or actions depending on conditions.
- In python, the following are conditional statements
 - if
 - if –else
 - if – elif –else

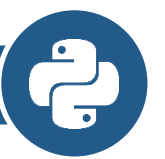


- if Statement
- if...else Statement
- if...elif...else Statement
- match-case Statement
- Nested if statements
- Ternary Operator



if Statement

- The if statement is used to perform operation on the basis of condition.
- The single if statement is used to execute the code if condition is true.
- **Syntax :-** if condition:
 statement(s)



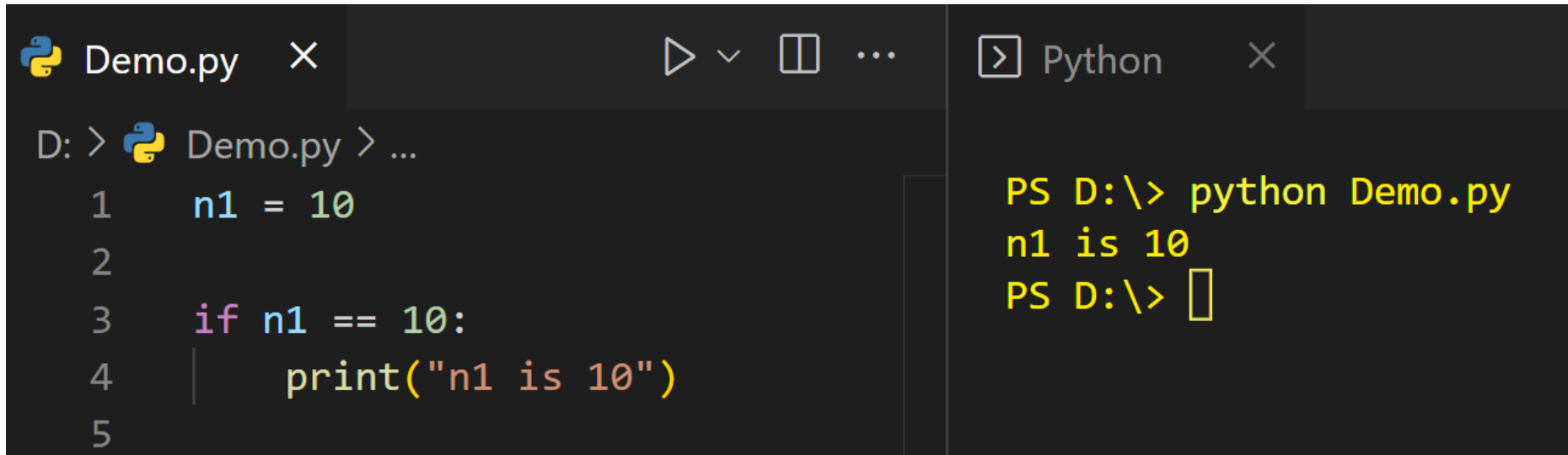
Indentation is required

- We can use any number of Space 1 or tab

```
hello.py 1 x
hello.py > ...
1 no1 = 10
2
3 if no1==10:
4     print("No1 is 10")
Expected indented block Pylance
```

Example 1

- Below program will check if a value is 10 or not (Comparison)
- == is used to check comparison of value



```
Demo.py x [Run] [Debug] [Tools] ... Python x
```

```
D: > python Demo.py > ...
```

```
1  n1 = 10
```

```
2
```

```
3  if n1 == 10:
```

```
4      print("n1 is 10")
```

```
5
```

```
PS D:\> python Demo.py
```

```
n1 is 10
```

```
PS D:\> 
```

```
n1 = 10
if n1 == 10:
    print("n1 is 10")
```

We can also use Brackets (Optional)

```
1  n1 = 10
2
3  if (n1 == 10):
4      |    print("N1 is 10")
5
```

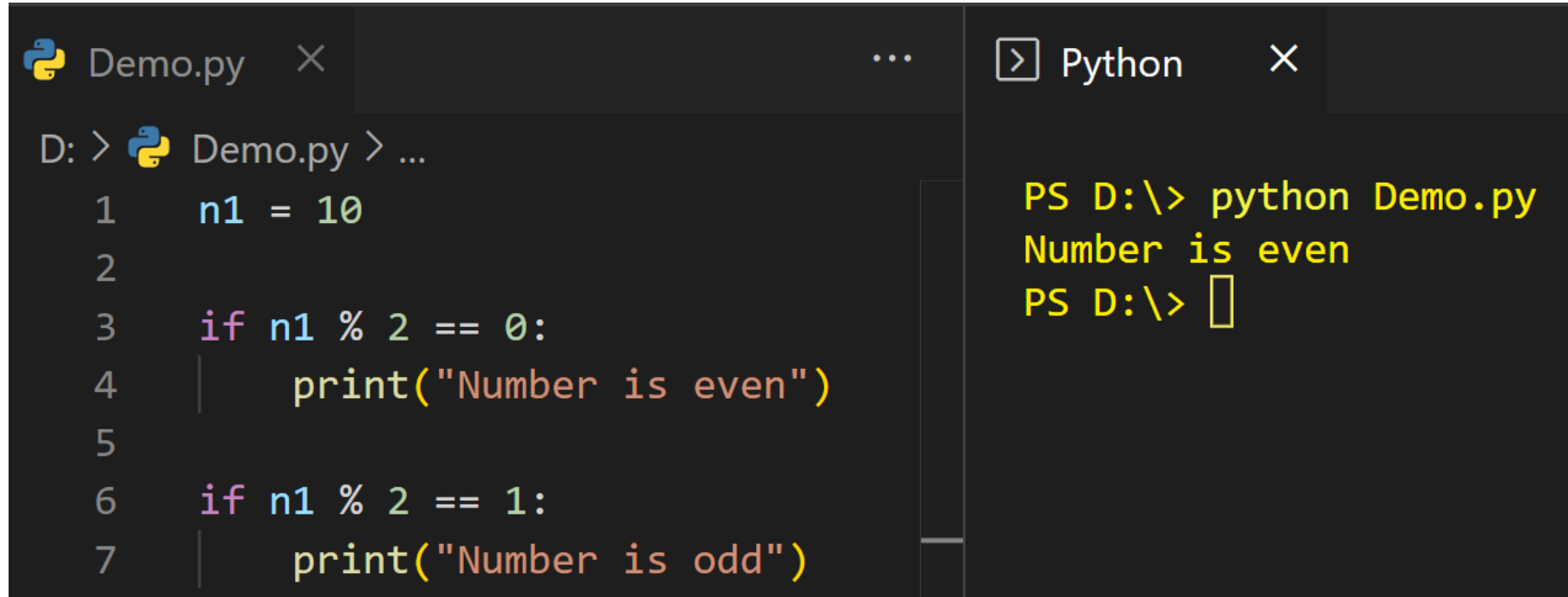
Example2

■ Even-Odd Program

```
n1 = 10

if n1 % 2 == 0:
    print("Number is even")

if n1 % 2 == 1:
    print("Number is odd")
```



The screenshot shows a Python IDE with two windows. The left window, titled 'Demo.py', contains the following code:

```
D: > Demo.py > ...
1  n1 = 10
2
3  if n1 % 2 == 0:
4      print("Number is even")
5
6  if n1 % 2 == 1:
7      print("Number is odd")
```

The right window, titled 'Python', shows the command prompt output:

```
PS D:\> python Demo.py
Number is even
PS D:\> 
```

```
1 a = 33
2 b = 200
3 if b > a:
4     print ("B is greater than A")
5 print ("Bye")
6
```

PROBLEMS

OUTPUT

DEBUG CONSOLE

PORTS

TERMINAL

```
D:\pythoncode>python demo.py
B is greater than A
Bye
```

if...else Statement

- The if-else statement provides an else block combined with the if statement which is executed in the false case of the condition.
- **Syntax :-**

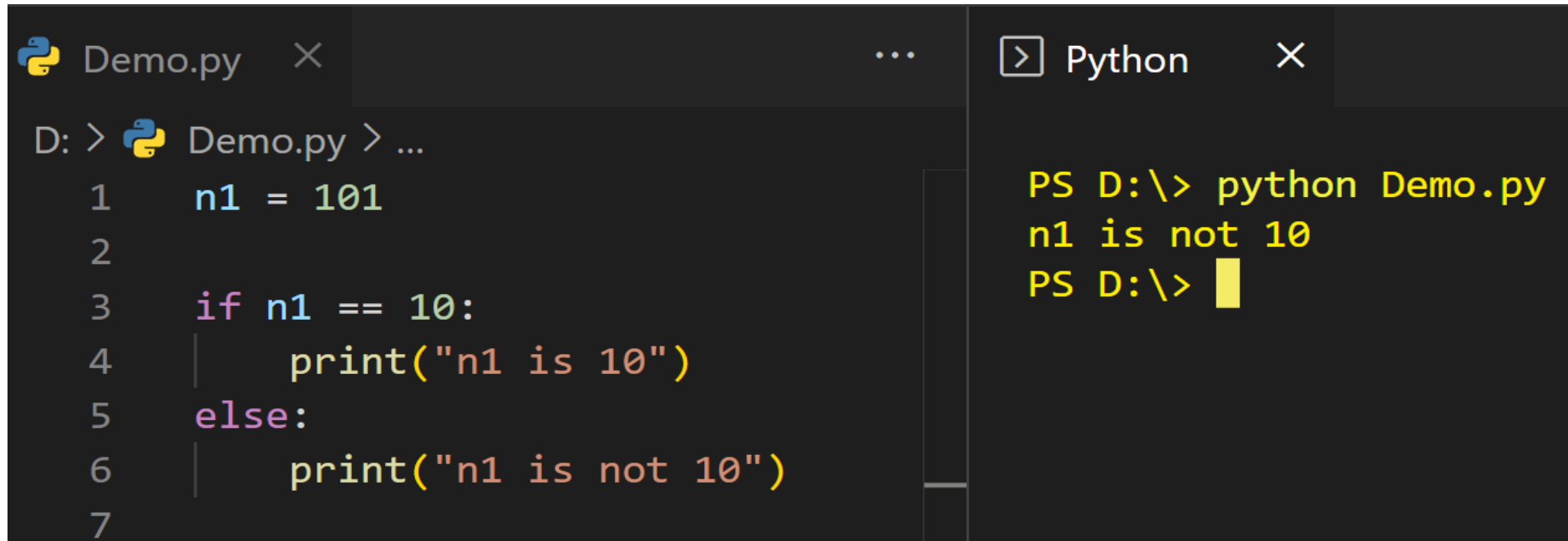
if condition:

Statements to be executed

else:

Statements to be executed

N1 is 10 or Not

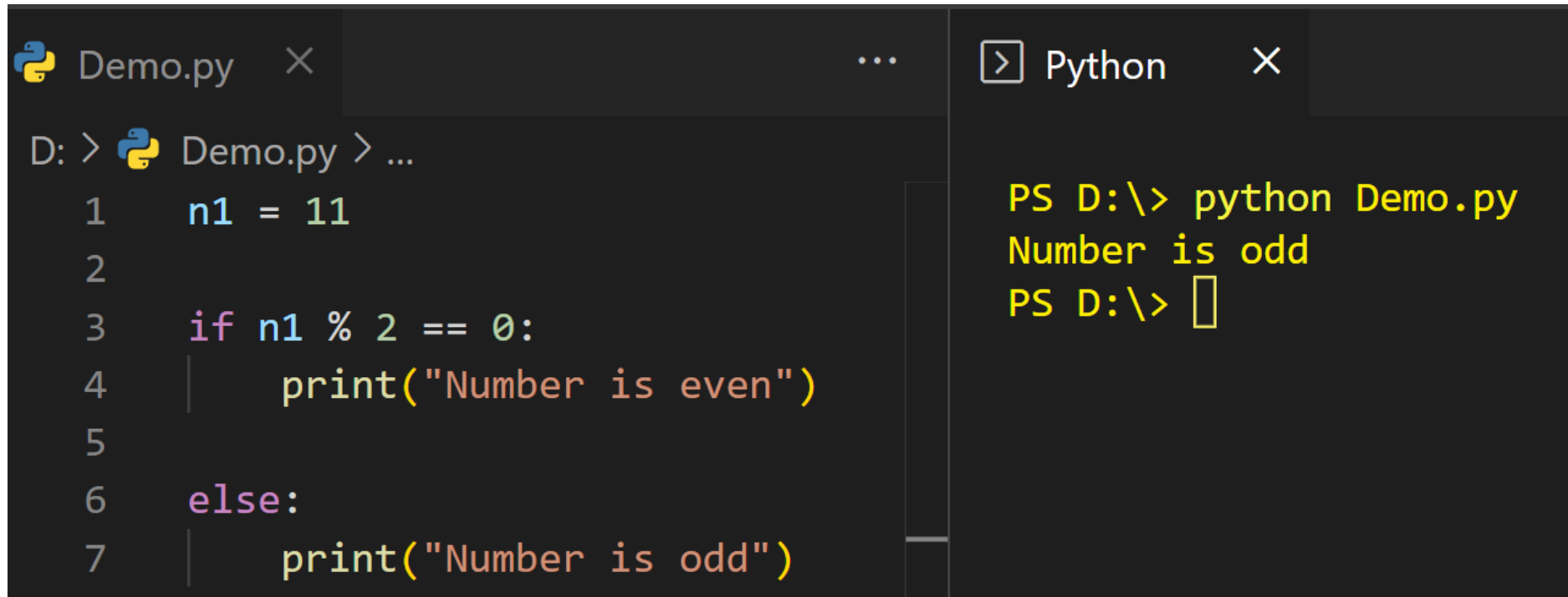


```
Demo.py × ... Python ×
D: > Demo.py > ...
1  n1 = 101
2
3  if n1 == 10:
4      print("n1 is 10")
5  else:
6      print("n1 is not 10")
7

PS D:\> python Demo.py
n1 is not 10
PS D:\> 
```

```
n1 = 101
if n1 == 10:
    print("n1 is 10")
else:
    print("n1 is not 10")
```

Even Odd Number

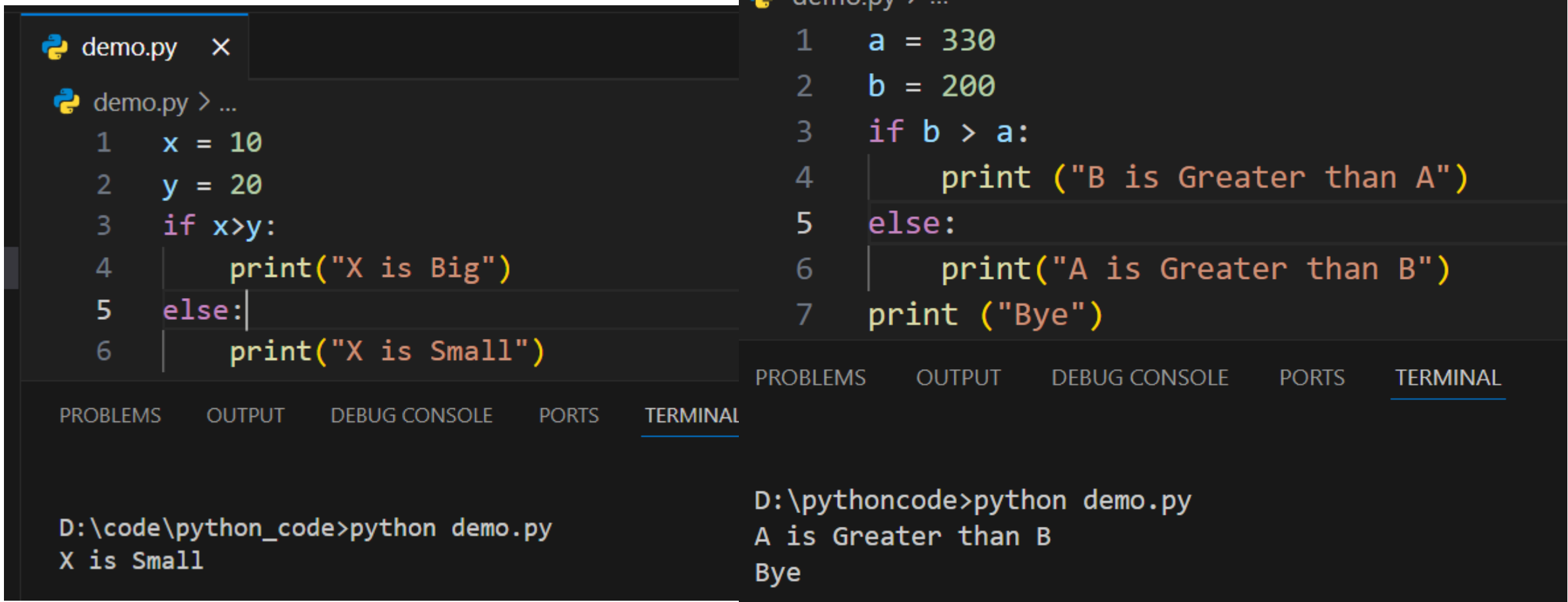


The image shows a code editor window with a file named 'Demo.py' and a terminal window. The code in 'Demo.py' is a Python script that checks if a number is even or odd. The terminal shows the command 'python Demo.py' being executed, resulting in the output 'Number is odd'.

```
Demo.py × ... Python ×  
D: > python Demo.py > ...  
1 n1 = 11  
2  
3 if n1 % 2 == 0:  
4     print("Number is even")  
5  
6 else:  
7     print("Number is odd")  
  
PS D:\> python Demo.py  
Number is odd  
PS D:\> 
```

```
n1 = 11  
if n1 % 2 == 0:  
    print("Number is even")  
else:  
    print("Number is odd")
```

X Y Find Bigger Number



The image shows a side-by-side comparison of two Python programs and their execution results. The left editor window shows a program that compares two variables, x and y, and prints a message based on which is larger. The right editor window shows a similar program but with different values for x and y, and it includes a 'Bye' message at the end.

```
demo.py x
demo.py > ...
1 x = 10
2 y = 20
3 if x>y:
4     print("X is Big")
5 else:
6     print("X is Small")

PROBLEMS OUTPUT DEBUG CONSOLE PORTS TERMINAL

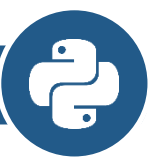
D:\code\python_code>python demo.py
X is Small
```

```
demo.py > ...
1 a = 330
2 b = 200
3 if b > a:
4     print ("B is Greater than A")
5 else:
6     print("A is Greater than B")
7 print ("Bye")

PROBLEMS OUTPUT DEBUG CONSOLE PORTS TERMINAL

D:\pythoncode>python demo.py
A is Greater than B
Bye
```

```
age=int(input("Enteryourage : ")) ifage>=18:  
print("Youareeligibleletovote!!") else:  
print("Sorry!youhavetowait!!"))
```



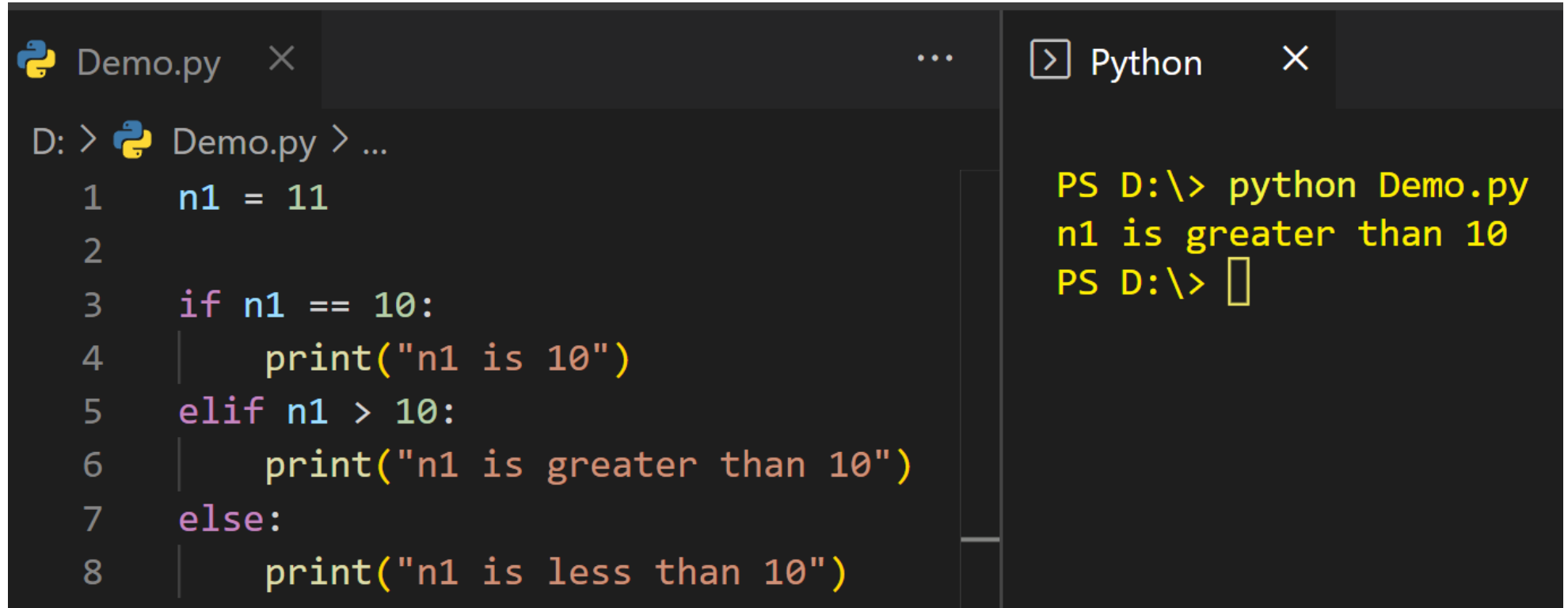
if...elif...else Statement

- The elif statement enables us to check multiple conditions and execute the specific block of statements depending upon the true condition among them
- **Syntax :-**
 - if condition:
 Statements to be executed
 - elif condition:
 Statements to be executed
 - elif condition:
 Statements to be executed
 - else:
 Statements to be executed

Example 1

```
n1 = 11

if n1 == 10:
    print("n1 is 10")
elif n1 > 10:
    print("n1 is greater than 10")
else:
    print("n1 is less than 10")
```



The screenshot shows a code editor window titled 'Demo.py' with the following Python code:

```
1  n1 = 11
2
3  if n1 == 10:
4      print("n1 is 10")
5  elif n1 > 10:
6      print("n1 is greater than 10")
7  else:
8      print("n1 is less than 10")
```

Below the code editor is a terminal window titled 'Python' showing the command prompt output:

```
PS D:\> python Demo.py
n1 is greater than 10
PS D:\> 
```

Take Values from User

```
hello.py x
hello.py > ...
1 no1 = int(input("Enter No:"))
2
3 if no1==10:
4     print("No1 is 10")
5 elif no1<10:
6     print("No1 is <10")
7 else:
8     print("No1 is >10")

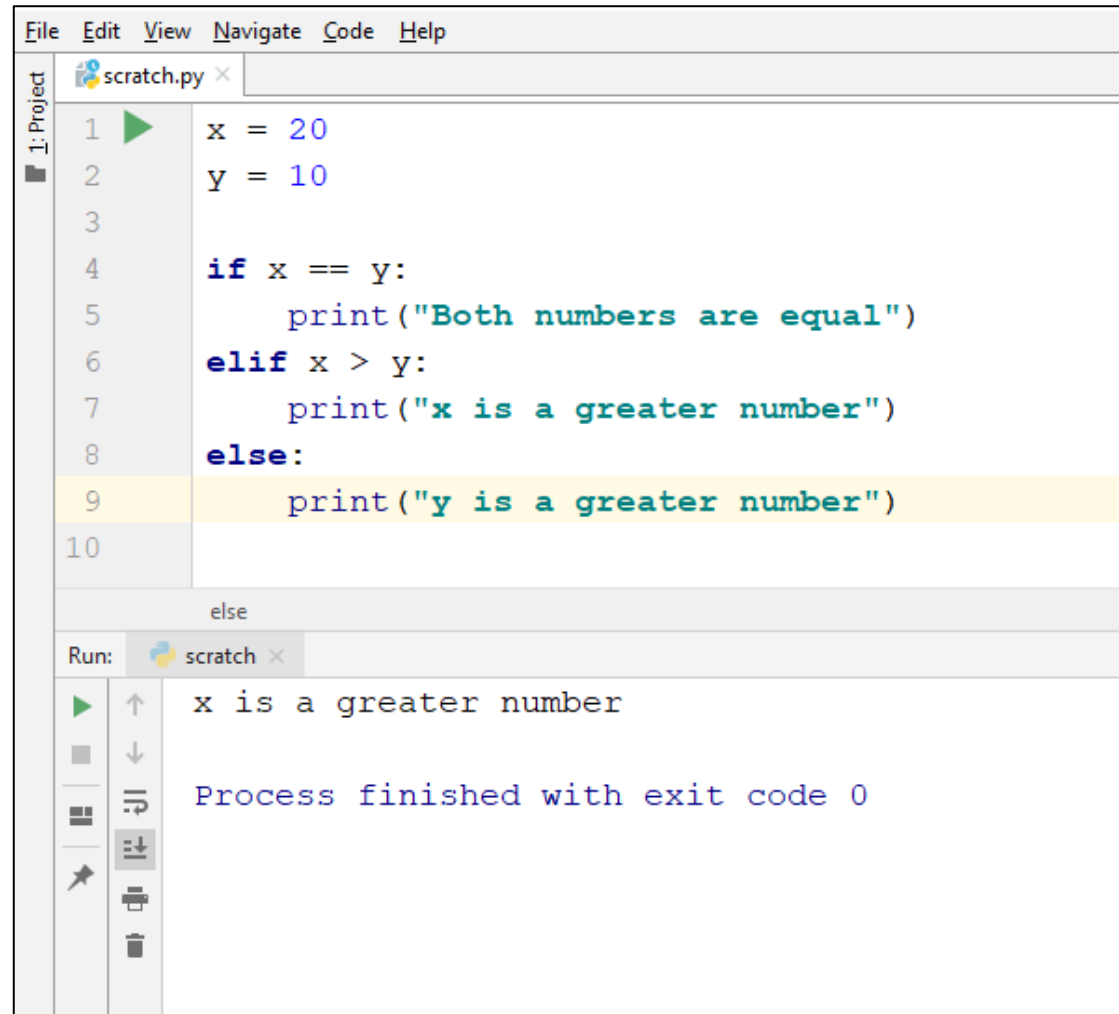
PROBLEMS OUTPUT DEBUG CONSOLE PORTS TERMINAL

D:\code\python_code>python hello.py
Enter No:20
No1 is >10
```

Example2

```
scratch.py x
1 n1 = -10
2 if n1 == 0:
3     print("n1 is 0")
4 elif n1 > 0:
5     print("n1 is positive")
6 else:
7     print("n1 is negative")
8
if n1 == 0
Run: scratch x
C:\Users\Devanshi\PycharmProjects\p
C:/Users/Devanshi/AppData/Roaming/
n1 is negative
```

Example3



```
File Edit View Navigate Code Help
scratch.py x
1: Project
1 x = 20
2 y = 10
3
4 if x == y:
5     print("Both numbers are equal")
6 elif x > y:
7     print("x is a greater number")
8 else:
9     print("y is a greater number")
10
else
Run: scratch x
x is a greater number
Process finished with exit code 0
```

Match



match-case Statement

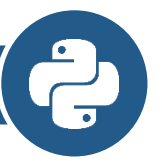
- The match-case statement is similar to switch statements
- Syntax:

```
match variable:  
    case matching_value_1: statement_1  
    case matching_value_2: statement_2  
    case matching_value_3: statement_3
```

Choice Based (Menu Driven)

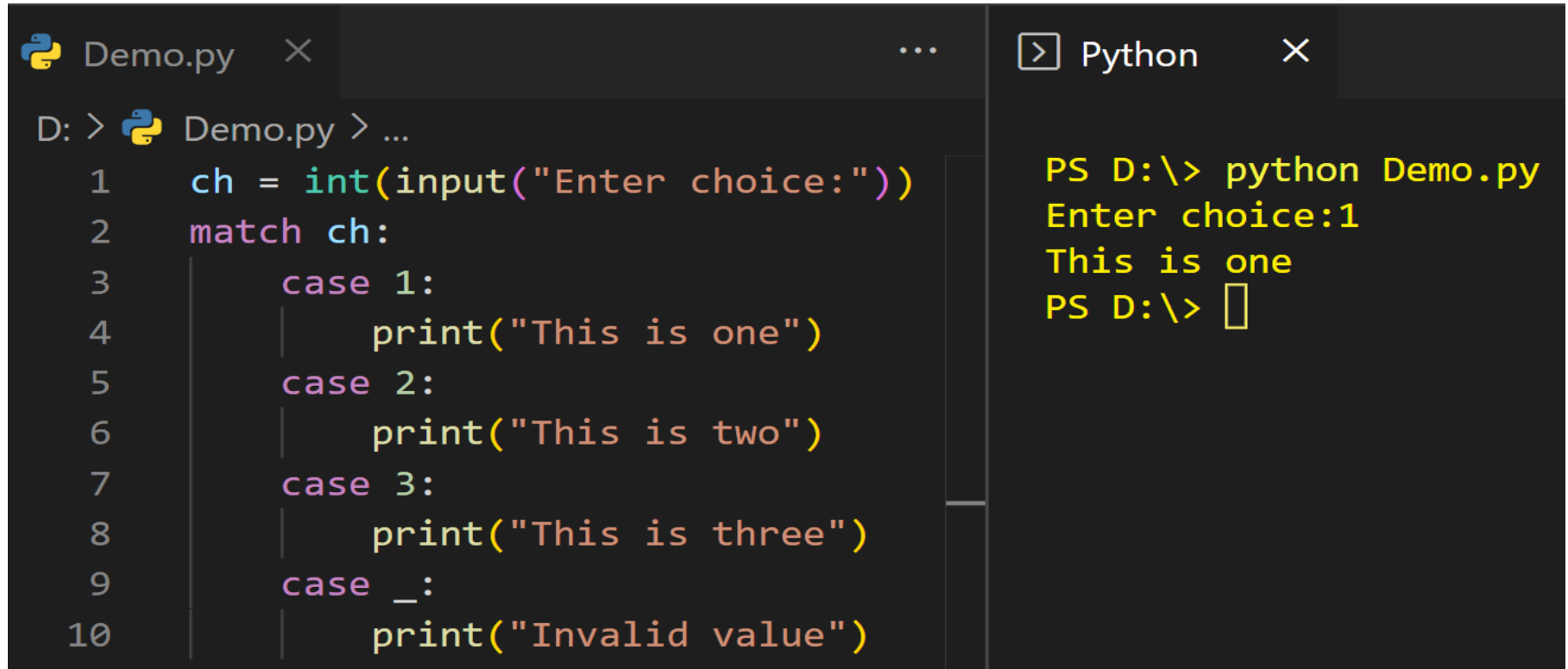
```
1  a = 10
2  match a:
3      case 1:
4          print("One")
5      case 2:
6          print("Two")
7      case 3:
8          print("Three")
9      case _:
10         print("Select 1-3")
```

Select 1-3



Example

```
ch = int(input("Enter choice:"))
match ch:
    case 1:
        print("This is one")
    case 2:
        print("This is two")
    case 3:
        print("This is three")
    case _:
        print("Invalid value")
```



The screenshot shows a code editor window titled 'Demo.py' with the following Python code:

```
D: > Demo.py > ...
1  ch = int(input("Enter choice:"))
2  match ch:
3      case 1:
4          print("This is one")
5      case 2:
6          print("This is two")
7      case 3:
8          print("This is three")
9      case _:
10         print("Invalid value")
```

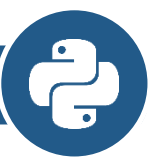
To the right, a terminal window titled 'Python' shows the execution output:

```
PS D:\> python Demo.py
Enter choice:1
This is one
PS D:\> 
```

```
hello.py > ...
1 choice = int(input("Enter No 1-5 :- "))
2
3 match choice:
4     case 1:
5         print("Monday")
6     case 2:
7         print("Tuesday")
8     case 3:
9         print("Wendsday")
10    case 4:
11        print("Thursday")
12    case 5:
13        print("Friday")
14    case _:
15        print("Weekend")
16
```

PROBLEMS OUTPUT DEBUG CONSOLE PORTS TERMINAL

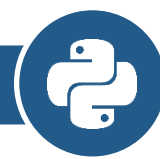
```
D:\code\python_code>python hello.py
Enter No 1-5 :- 6
Weekend
```



Choice Based

```
1 no1 = 10
2 no2 = 20
3 op = "*"
4 match op:
5     case "+":
6         c = no1 + no2
7         print("Sum is ",c)
8     case "-":
9         c = no1 - no2
10        print("Sum is ",c)
11    case "*":
12        c = no1 * no2
13        print("Multiplication is ",c)
14    case "/":
15        c = no1 + no2
16        print("Divison is ",c)
17    case _:
18        print("select + - * /")
19
```

```
D:\code\python_code>python matchdemo.py
Multiplication is 200
```

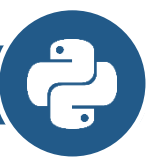


Ternary Operator

- Ternary operators are also known as **conditional expressions**.
- They are operators that evaluate something based on a condition being true or false.
- It is shortened way of writing an if-else statement.

- **Syntax :**

`[on_true] if [expression] else [on_false]`



Example

```
Demo.py ×
D: > Demo.py > ...
1 a, b = 10, 20
2 ans = a if a < b else b
3 print("Smallest value is ", ans)
4
```

```
a, b = 10, 20
ans = a if a < b else b
print("Smallest value is ", ans)
```

```
Python ×
PS D:\> python Demo.py
Smallest value is 10
PS D:\> 
```

```
ifdemo1.py > ...
1 no1 = 100
2 ans = "True" if no1<10 else "False"
3 print(ans)
```

PROBLEMS OUTPUT DEBUG CONSOLE PORTS TERMINAL

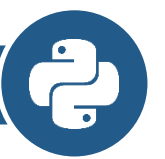
```
D:\code\python_code>python ifdemo1.py
False
```



- An if statements inside if statements, this is called a nested if.

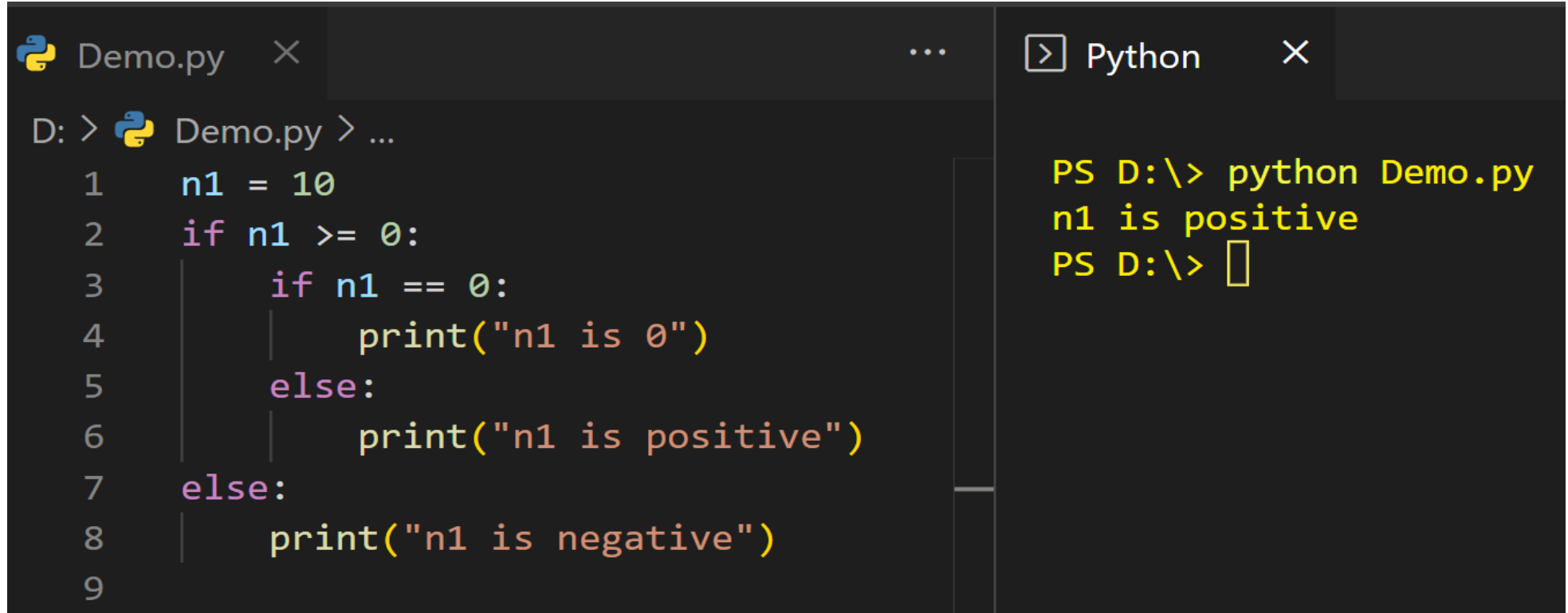
- **Syntax :-**

```
if condition:
    if condition:
        Statements to be executed
    else:
        Statements to be executed
else:
    if condition:
        Statements to be executed
    else:
        Statements to be executed
```



Nested If (If Inside IF)

```
n1 = 10
if n1 >= 0:
    if n1 == 0:
        print("n1 is 0")
    else:
        print("n1 is positive")
else:
    print("n1 is negative")
```



```
Demo.py × ... Python ×
D: > Demo.py > ...
1  n1 = 10
2  if n1 >= 0:
3      if n1 == 0:
4          print("n1 is 0")
5      else:
6          print("n1 is positive")
7  else:
8      print("n1 is negative")
9

PS D:\> python Demo.py
n1 is positive
PS D:\> 
```

Example 2

```
main.py x
1  n1 = -10
2  if n1 == 0:
3      print("n1 is 0")
4  else:
5      if n1 > 0:
6          print("n1 is positive")
7      else:
8          print("n1 is negative")
9
Run: main x
C:\Users\Devanshi\PycharmProjects\
C:/Users/Devanshi/PycharmProjects
n1 is negative
```

```
1  n1 = 11
2  if n1>0:
3      if n1==10:
4          print("N1 is 10")
5      elif n1>10:
6          print("N1 is >10")
7      else:
8          print("N1 is < 10")
9  else:
10     print("Enter Positive Value")
```

```
n1 = 11
if n1>0:
    if n1==10:
        print("N1 is 10")
    elif n1>10:
        print("N1 is >10")
    else:
        print("N1 is < 10")
else:
    print("Enter Positive Value")
```

- Take 3 numbers from User and Check Which Number is Greatest.
- Take a 5 Subject Marks from users Print All Subject Total, Percentage and Make a Proper Grade system using if elif.
- Check in Each Subject marks is <18 then do not print grade just print fail message.

Let's Connect



Akash Padhiyar

Tech Educator • Consultant Tech
Mentor • Founder

On a Mission 🚀 to Bridge the
Gap Between Campus & Corporate

Akashsir.com

SOCIAL MEDIA

#DailyUpdates



@akashsirdotcom

#FreeCode



@akashpadhiyar

#TechUpdates



@akashpadhiyar

#FreeTutorials



@akashtecholabs

#HelpDesk



+91 99786 21654

WEBSITES

> **www.akashtecholabs.com**

> **www.akashsir.com**



akash@akashtecholabs.com

📍 202, Arya Arcade, Mithakhali Six Road, Navrangpura, Ahmedabad, Gujarat 380009